

Problem 4.15 on composition type

Exact negative answer in later literature

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Status. `literature_already_answered`. This packet identifies and checks a counterexample already published in later literature; it does not claim a new result.

Source problem

Turco and Villafa e, *Galois connection between Lipschitz and linear operator ideals and minimal Lipschitz operator ideals*, arXiv:1807.10786, ask in Problem 4.15:

If a Banach Lipschitz operator ideal $\mathcal{I}$ has $\mathcal{I}^{\min}$ or $\mathcal{I}^{\max}$ of composition type, must $\mathcal{I}$ itself be of composition type?

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As a consequence of Proposition 4.13, every Banach Lipschitz ideal belonging to the sublattice $\mathfrak{J}_{\mathcal{N}_p^L}$ must be not of composition type. Indeed, if $\mathcal{A} \circ \text{Lip}_0 \in \mathfrak{J}_{\mathcal{N}_p^L}$, then $\mathfrak{J}_{\mathcal{N}_p^L} = \mathfrak{J}_{\mathcal{A} \circ \text{Lip}_0}$ which implies that $\mathcal{A}^{\min} \circ \text{Lip}_0 = (\mathcal{N}_p^L)^{\min}$. In general, we have.

Proposition 4.14. *Let $\mathcal{I}$ be a Banach Lipschitz operator ideal such that $\mathcal{I}^{\min}$ or $\mathcal{I}^{\max}$ is not of composition type. Then, every Banach Lipschitz operator ideal $\mathcal{J} \in \mathfrak{J}_{\mathcal{I}}$ is not of composition type. In particular, $\mathcal{I}$ is not of composition type.*

We do not know if the converse of the above proposition holds.

Problem 4.15. *Let $\mathcal{I}$ be a Banach Lipschitz operator ideal. Suppose that $\mathcal{I}^{\min}$ or $\mathcal{I}^{\max}$ is of composition type. Does $\mathcal{I}$ must be of composition type?*

REFERENCES

- [1] D. Achour, P. Rueda, E. A. S nchez-P rez, and R. Yahi, *Lipschitz operator ideals and the approximation property*, J. Math. Anal. Appl. **436** (2016), no. 1, 217–236. MR 3440091
- [2] R.F. Arens and J. Eells, Jr., *On embedding uniform and topological spaces*, Pacific J. Math.

Exact later answer

Albarrac n and Turco, *On the Lipschitz operator ideal $\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0$* , arXiv:2211.14240v2, explicitly cite Problem 4.15 and give a negative answer. Let $\overline{\mathcal{F}}$ denote the Banach ideal of approximable linear operators and set

$$\mathcal{I} = \text{Lip}_0 \circ \overline{\mathcal{F}} \circ \text{Lip}_0.$$

Their Propositions 2.13 and 2.16 prove, respectively,

$$\mathcal{I} \text{ is not of composition type,} \quad \mathcal{I}^{\min} = \overline{\mathcal{F}} \circ \text{Lip}_0 \text{ is of composition type.}$$

Thus $\mathcal{I}$ satisfies the premise of Problem 4.15 through its minimal kernel, but violates the proposed conclusion. A counterexample for the minimal alternative is enough to make the disjunctive implication false; no statement about $\mathcal{I}^{\max}$ is needed.

As a consequence, we have the following

Corollary 2.12. *Let $\mathcal{A}$ be a Banach operator ideal. The Lipschitz operator ideal $\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0$ is not minimal.*

In particular, $\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0$ does not coincide with $(\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0)^{\min}$ or $\mathcal{A}^{\min} \circ \text{Lip}_0$, since $\mathcal{A}^{\min} \circ \text{Lip}_0$ is a minimal Lipschitz operator ideal [30, Theorem 4.8]. Almost with the same proof of Proposition 2.11 we obtain the following.

Proposition 2.13. *Let $\mathcal{A}$ be a Banach operator ideal. The Lipschitz operator ideal $\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0$ is not of composition type*

Proof. If $\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0$ were of composition type, then by [30, Proposition 3.16], we would have

$$((\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0) \cap \mathcal{L}) \circ \text{Lip}_0 = \text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0.$$

In particular, for $E \neq \{0\}$ a finite-dimensional Banach space, since $\delta_E \in \text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0(E, \mathcal{A}(E))$, we get that its linearization, $\text{id}_{\mathcal{A}(E)}$, belongs to $(\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0) \cap \mathcal{L}(\mathcal{A}(E), \mathcal{A}(E))$. By Theorem 2.4, we have that $\text{id}_{\mathcal{A}(E)} \in (\mathcal{A}^{\min})^{\text{reg}}(\mathcal{A}(E), \mathcal{A}(E))$ and,

How the two supporting propositions work

Proposition 2.13 is more general: for every Banach linear operator ideal $\mathcal{A}$, the strong Lipschitz ideal $\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0$ is not of composition type. For a nonzero finite-dimensional Banach space E , this ideal contains the Dirac map $\delta_E : E \rightarrow \mathcal{F}(E)$. Were the ideal of composition type, its standard linearization criterion would put $I_{\mathcal{F}(E)}$ in $(\mathcal{A}^{\min})^{\text{reg}}$. That identity would be compact, although the Lipschitz-free space $\mathcal{F}(E)$ is infinite dimensional, a contradiction. Taking $\mathcal{A} = \overline{\mathcal{F}} = \overline{\mathcal{F}}^{\min}$ gives the first displayed assertion about $\mathcal{I}$.

Proposition 2.16 treats every closed Banach operator ideal $\mathcal{A}$. Minimality gives

$$(\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0)^{\min} \subseteq \overline{\mathcal{F}} \circ \text{Lip}_0,$$

while $\mathcal{A} \circ \text{Lip}_0 \subseteq \text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0$ gives

$$\mathcal{A}^{\min} \circ \text{Lip}_0 \subseteq (\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0)^{\min}.$$

For closed $\mathcal{A}$, one has $\mathcal{A}^{\min} = \overline{\mathcal{F}}$, so the two inclusions force equality with $\overline{\mathcal{F}} \circ \text{Lip}_0$, a composition ideal. The later paper then specializes to $\mathcal{A} = \overline{\mathcal{F}}$ and explicitly labels the result a negative answer to Problem 4.15.

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Proof. Since $(\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0)^{\min}$ is minimal, by [30, Proposition 4.5] we have that $(\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0)^{\min} \subset \overline{\mathcal{F}} \circ \text{Lip}_0$. On the other hand, we have $\mathcal{A} \circ \text{Lip}_0 \subset \text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0$, which implies that

$$\mathcal{A}^{\min} \circ \text{Lip}_0 \subset (\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0)^{\min}.$$

Since $\mathcal{A}$ is closed, we know that $\mathcal{A}^{\min} = \overline{\mathcal{F}}$, and the proof is complete. $\square$

In particular, for the ideal of approximable operators, by Proposition 2.13, $\text{Lip}_0 \circ \overline{\mathcal{F}} \circ \text{Lip}_0$ is not of composition type, but by Proposition 2.16 its minimal kernel is of composition type, giving a negative answer of [30, Problem 4.15].

We do not know any other example besides $\overline{\mathcal{F}}$ of a minimal Banach operator ideal such that $(\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0)^{\min}$ is of composition type. However, we have the following result. Before stating it, recall that, following [26, Definition 4.7.1], an operator $T: E \rightarrow F$ belongs to the *surjective hull* of a Banach operator ideal $\mathcal{A}$, denoted as $\mathcal{A}^{\text{sur}}$, if $T \circ Q_E \in \mathcal{A}(\ell_1(B_E), F)$ where $Q_E: \ell_1(B_E) \rightarrow E$ is the canonical quotient map.

Proposition 4.15. *Let $\mathcal{A}$ be a Banach operator ideal. Then $(\text{Lip}_0 \circ \mathcal{A}^{\min} \circ \text{Lip}_0)^{\min} = \overline{\mathcal{F}} \circ \text{Lip}_0$ if and only if $\mathcal{A}^{\min} = \mathcal{A}^{\text{sur}}$.*

Proof intuition

The original implication fails because taking the minimal kernel can erase the nonlinear “two-sided Lipschitz” obstruction. The original ideal is large enough to contain Dirac maps whose linearizations are identities on infinite-dimensional free spaces, preventing composition type. Its minimal kernel, however, collapses exactly to the ordinary approximable composition ideal $\overline{\mathcal{F}} \circ \text{Lip}_0$.

Exactness, scope, and provenance

This is a direct answer, not an inference: the supporting authors cite the source as reference [30], name Problem 4.15, and state that their example gives its negative answer. It settles the yes/no problem completely, while their following sentence leaves open only the classification of further examples besides $\overline{\mathcal{F}}$.

The target PDF was compiled without source edits from the run’s archived arXiv source. The supporting PDF was reused from the already verified arXiv:2211.14240 packet in this run. Exact page crops of the question, noncomposition proposition, and explicit negative-answer passage are bundled above.

References. [1] P. Turco and R. Villafañe, *Galois connection between Lipschitz and linear operator ideals and minimal Lipschitz operator ideals*, J. Funct. Anal. 277 (2019), 434–451; arXiv:1807.10786, Problem 4.15. [2] N. Albarracín and P. Turco, *On the Lipschitz operator ideal $\text{Lip}_0 \circ \mathcal{A} \circ \text{Lip}_0$* , arXiv:2211.14240v2 (2023), Propositions 2.13 and 2.16.